

System-Level Modeling of Multi-Accelerator Architectures via Analytical Cost Model Integration

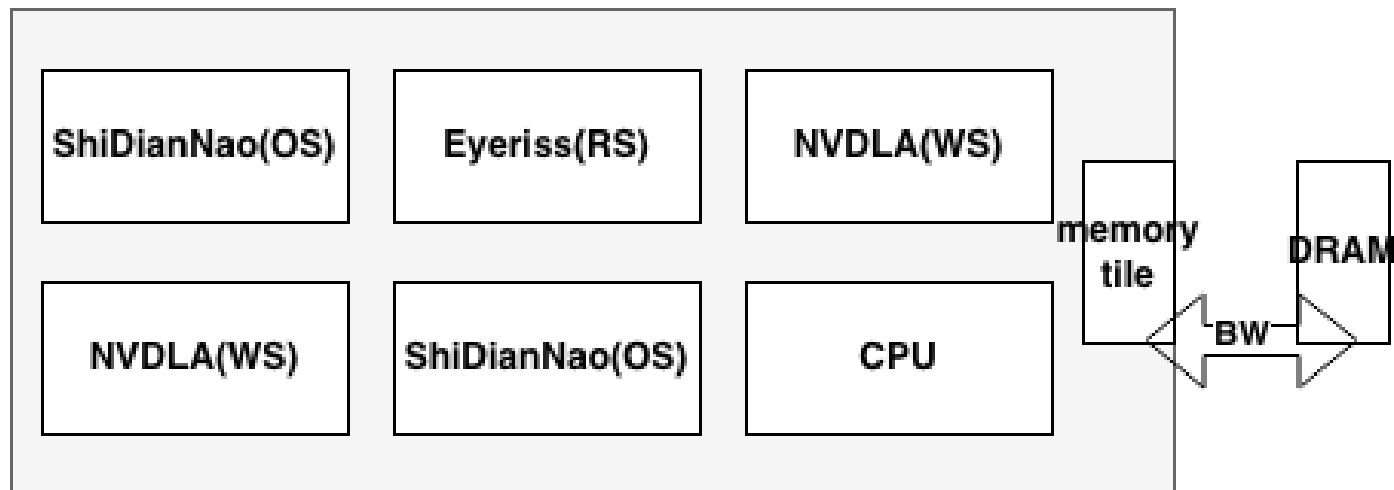
COMS E6868 – Embedded Scalable Platforms – Spring 2025

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Goal

- **Unify accelerator-level and system-level modeling**
- Use analytical cost models (**MAESTRO**, **Timeloop**) for individual accelerators
- Add explicit modeling of **NoC** and **shared DRAM** for multi-accelerator systems



Midterm focus

- Study how each accelerator's dataflow can be expressed in each framework:
- **ShiDianNao** (output-stationary, OS)
- **NVDLA** (weight-stationary, WS)
- **Eyeriss** (row-stationary, RS)
- Same layer, same PE count, in **both** MAESTRO and Timeloop → compare latency, energy, EDP.

Describe hardware the way each tool expects

MAESTRO

- **HW file** (.m): num_pes, L1/L2 size constraints, NoC/off-chip BW
- **Mapping file** (.m): Dataflow (TemporalMap, SpatialMap, Cluster over K,C,R,S,Y,X)

Timeloop

- **Architecture YAML**: hierarchy (MainMemory → PE array → Buffer → MACC), technology, global_cycle_seconds, spatial mesh (e.g. $16 \times 16 = 256$ PEs)
- **Mapping YAML**: temporal/spatial factors

Convert MAC to μJ

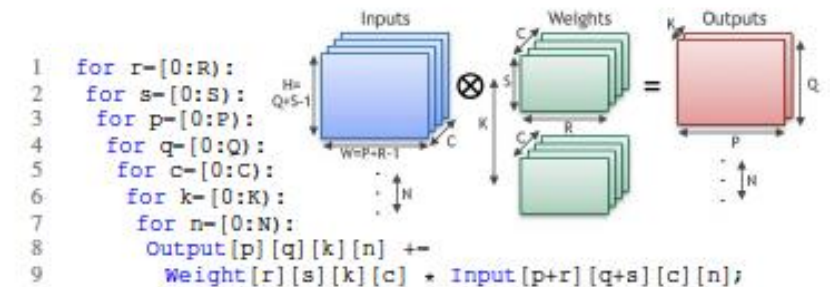
- **MAESTRO** reports energy in “**X MAC energy**” (abstract units)
- **Timeloop** reports energy in μJ (from Accelergy/ERT)
- To compare: **energy (μJ) = (X MAC energy) $\times$ 0.21 pJ/MAC $\times 10^{-6}$**
- We use **0.21 pJ/MAC** so MAESTRO and Timeloop sit in the same scale for plots

Quantity	MAESTRO	Timeloop
Latency	Runtime: N cycles	Cycles: N (cycle = e.g. 1 ns)
Energy	"X MAC energy" $\rightarrow$ convert to μJ	μJ
Utilization	% (from avg. utilized PEs)	% of PEs busy
EDP	cycles * energy	Cycles * energy

Same layer, three accelerators (MAESTRO + Timeloop)

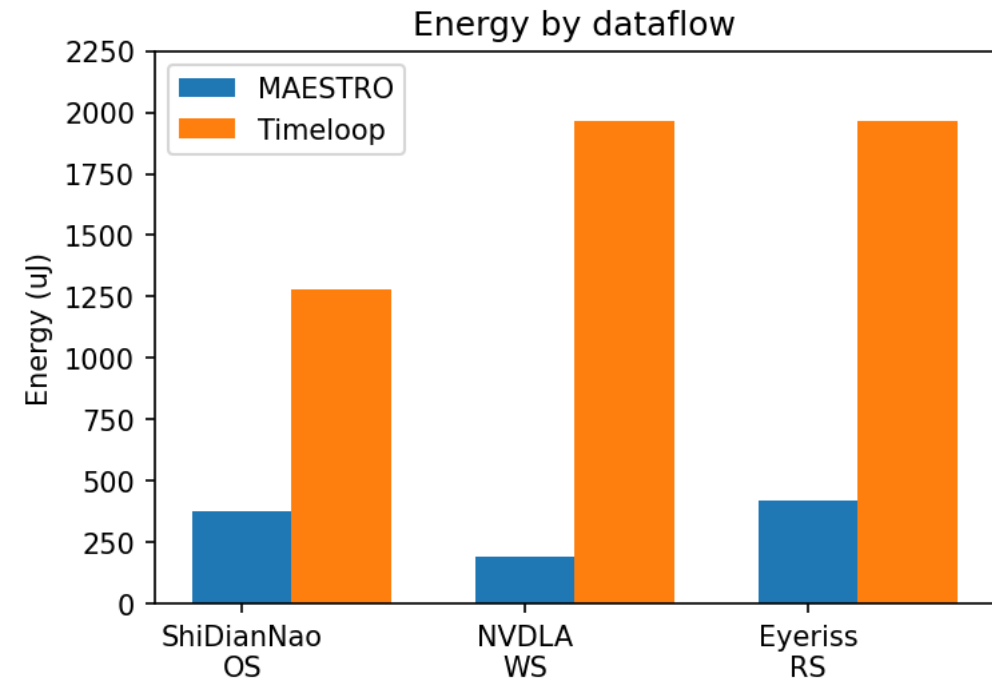
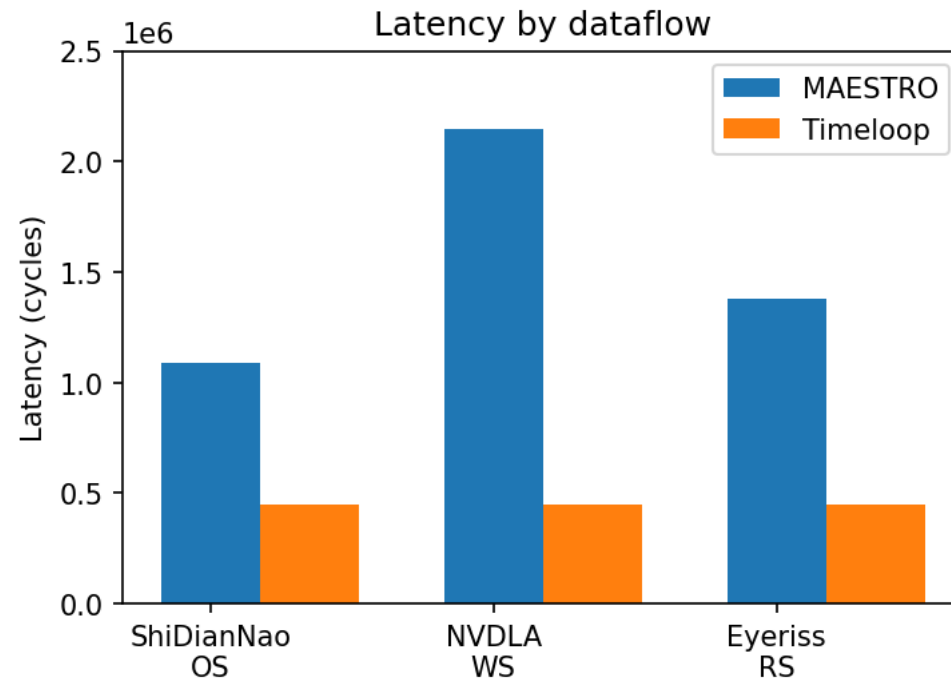
- **Same layer (all runs):** $K=C=64$, $R=S=3$, $P=Q=56 \rightarrow$ **107M MACs**
Same accelerator size: 256 PEs in both tools

Dataflow	MAESTRO mapping	Timeloop
OS	ShiDianNao (output-stationary)	Same-layer OS mapping
WS	NVDLA (weight-stationary)	Same-layer WS mapping
RS	Eyeriss (row-stationary)	Eyeriss mapper



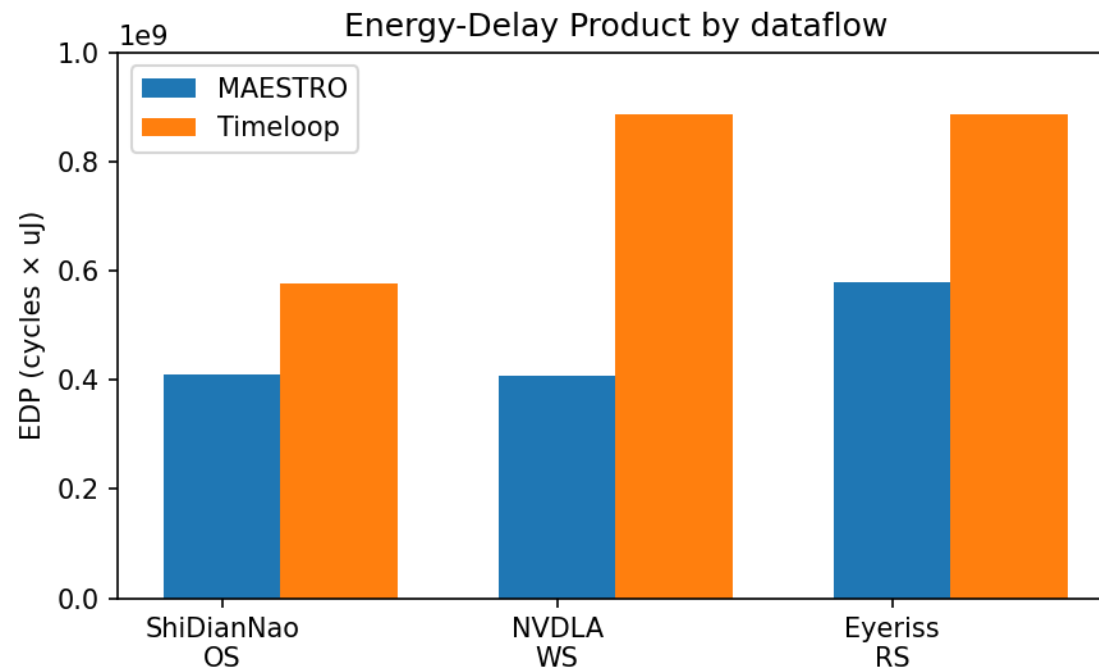
Results

- **Latency:** MAESTRO $\sim 1\text{--}2\text{M}$ cycles; Timeloop $\sim 452\text{K}$ (256-PE equivalent). MAESTRO models a real 256-PE execution, while Timeloop is a scaled-down 1-PE result.
- **Energy:** MAESTRO $\sim 190\text{--}420\text{ }\mu\text{J}$; Timeloop $\sim 1280\text{ }\mu\text{J}$. Timeloop's energy is higher because it uses Accelergy and models a full hierarchy(MainMemory, Buffer, MACC) with energy per read/write.



Results

- MAESTRO and Timeloop use **different** input formats, cycle models, and energy models
- Raw **latency** and **energy** can each be higher or lower in one tool
- EDP = latency × energy** captures the tradeoff; the product tends to be **more comparable** across tools



Next steps

- Add NoC and shared DRAM models (by Mar. 18)
- **Integration of inter-accelerator communication and DRAM access modeling** (by Apr. 1)
- **Simulation of representative DNN workloads** on multi-accelerator systems (by Apr. 15)
- **System-level optimization** (e.g. inspired by MAGMA, SCAR) (by May 4)
- **Final report and presentation** (May 14)